

Spatially Coupled Convolutional Codes via Block-Oriented Accumulation: Analysis and Quantized Decoding

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Abstract—Construction of high-performance spatially coupled codes has received considerable attention recently. This paper presents a new construction of spatially coupled convolutional codes (SC-CC), in which the codewords of the underlying convolutional codes are coupled together with the block-oriented accumulator (BA). The resulting codes are referred to as the BA-based SC-CCs (BA-SC-CCs). First, we employ the extrinsic information transfer (EXIT) chart analysis to study the iterative decoding thresholds of BA-SC-CCs. Our analysis indicates that, with appropriate component convolutional codes, the iterative decoding thresholds of BA-SC-CCs can be within 0.1 dB of the channel capacities. Second, we prove that the free distance of a BA-SC-CC is greater than that of the underlying convolutional code. Third, we present a method to analyze the decoding error propagation phenomenon of BA-SC-CCs. To enhance the decoding performance, we finally introduce a quantized threshold-based Max-Log-MAP (T-Max-Log-MAP) decoder for CCs, which can be incorporated into the decoder of BA-SC-CCs. Extensive numerical results demonstrate the advantages of the proposed codes and the new quantized decoder. Specifically, terminated BA-SC-CCs perform comparably to the braided convolutional codes (BCC) and outperform regular spatially coupled low-density parity-check (SC-LDPC) codes and spatially coupled repeater-convolutional-combiner (SC-RCC) codes. Furthermore, the proposed quantized decoder performs better than the conventional quantized decoder.

Index Terms—Block-oriented accumulator, convolutional codes, spatial coupling.

I. INTRODUCTION

CONSTRUCTION and analysis of high-performance spatially coupled codes for merging applications has received considerable attention [1], [2], [3]. Spatially coupled codes are constructed by interconnecting a series of disjoint graph-based codes into a single graph-based code.

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The first class of spatially coupled codes is that of spatially coupled low-density parity-check (SC-LDPC) codes [1], [4], [5], [6], [7], [8]. Analytical results showed that the iterative decoding threshold of an SC-LDPC ensemble achieves the maximum *a posteriori* (MAP) threshold of the underlying LDPC block ensemble over binary-input memoryless output-symmetric channels [4], [9]. This phenomenon, known as *threshold saturation*, makes spatial coupling an attractive technique for constructing capacity-achieving codes.

Since the advent of SC-LDPC codes, various types of spatially coupled codes have been introduced, including spatially coupled turbo-like (SC-TL) codes [10], [11], [12], block Markov superposition transmission (BMST) [13], [14], [15], and spatially coupled repeater-convolutional-combiner (SC-RCC) codes [16]. These works typically focused on illustrating the merits of spatial coupling in asymptotic latency regime.

In addition to theoretical advancements, significant efforts have been made to construct high-performance SC-LDPC codes. Spatially coupled repeat-accumulate (SC-RA) codes were studied in [17]. Algebraic constructions of time-invariant SC-LDPC codes were presented in [7]. In [18], the authors proposed to design structured SC-LDPC codes by maximizing the girth and minimizing the number of short cycles. In [19], the authors studied the short cycles of periodically time-varying SC-LDPC codes for improved code construction. The construction of short-length SC-LDPC codes by eliminating small trapping sets was investigated in [20]. Constructions of rate-compatible LDPC and SC-LDPC codes were investigated in [21], [22], and [23]. Graph extension was used in [21] to construct rate-compatible LDPC codes. Results in [22] showed that iterative decoding thresholds of randomly punctured SC-LDPC codes on additive white Gaussian noise (AWGN) channel can be estimated from the iterative decoding threshold of the un-punctured code and the design rate. Edge reallocation was used in [23] to construct high-performance rate-compatible SC-LDPC codes. Algebraic constructions of globally coupled LDPC codes with tail-biting structure were proposed in [24]. In [25], applications of SC-LDPC codes in joint source-channel coding were investigated. A novel technique based on partial spatial coupling was introduced in [26] to reduce the rate loss of SC-LDPC codes. The recently introduced free-ride codes are used in [27] to construct globally coupled LDPC codes.

More recently, attention has shifted to constructing spatially coupled codes based on strong basic codes, including turbo code, Hadamard code, convolutional codes, polar codes, and algebraic codes. The resulting codes include spatially coupled product-like codes [28], [29], [30], [31], SC-TL codes [10], [32], partially coupled codes [33], [34], [35], and spatially coupled generalized LDPC codes [36], [37], [38]. This paper is intended to construct high-performance terminated spatially coupled codes via coupling strong basic codes.

Designing low-complexity decoding algorithms for high-performance spatially coupled codes based on strong component codes is of practical interest, especially when these codes are used in applications with stringent complexity constraints. The min-sum algorithm (MSA) is the first class of low-complexity decoder for LDPC codes. Since the invention of MSA, various low-complexity decoders had been proposed in the literature, including attenuated MSA (AMSA) [39], weight bit-flip decoder [40], reliability-based decoder [41], and finite alphabet iterative decoder [42]. To lower the error floor of quantized MSA, a threshold-based attenuated MSA (TAMSA) was proposed in [43]. In [44], Ghassemi showed that non-uniform quantization performs better than uniform quantization. Based on the principle of maximizing mutual information (MMI), non-uniform quantization was designed for LDPC codes in [45]. A novel decoding paradigm for LDPC codes based on reconstruction-computation-quantization was proposed in [46].

Analyzing the free distance of spatially coupled codes remains a significant challenge [3], [47]. While existence-type lower bounds had been established for SC-LDPC codes with large constraint lengths [48], further results in [49] extended these bounds to periodically time-varying SC-LDPC codes. Asymptotic analyses have also shown that the free distance of SC-LDPC codes constructed from protograph-based LDPC block codes increases linearly with the constraint length [50]. More recently, new upper bounds on the free distance for periodically time-varying SC-LDPC codes were derived in [51], highlighting their advantage over their time-invariant counterparts. However, for SC-LDPC codes and SC-TL codes, the only general lower bound currently known is that spatial coupling does not reduce the minimum distance of the underlying uncoupled codes [47], [51]. These observations motivate further investigation into the free distance properties of terminated spatially coupled codes.

In this paper, we propose to use block-oriented accumulator (BA) to construct high-performance terminated spatially coupled convolutional codes (SC-CCs). The resulting codes are referred to as BA-SC-CCs. In BA-SC-CCs, the codewords of the underlying convolutional code (CC) are taken as the input of a block-oriented accumulator, which introduces memory among these codewords. The main contributions of this paper are summarized as follows.

- First, the convergence behaviors of BA-SC-CCs can be analyzed by the extrinsic information transfer (EXIT) chart. The analysis shows that, with appropriate component convolutional code, the thresholds of BA-SC-CCs can be within 0.1 dB of the channel capacity.

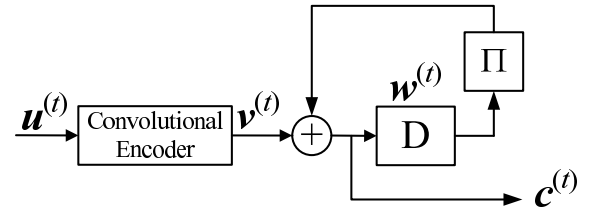


Fig. 1. Encoder of a BA-SC-CC, where the node D is a length- N register and Π is an interleaver of size N .

- Second, we prove that the free distance of the proposed BA-SC-CC is about three times as large as the free distance of the underlying convolutional code. To the best of our knowledge, this is the first explicit and nontrivial lower bound reported for terminated spatially coupled codes.
- Third, we introduce the concept of the *self-recovery threshold* (SRT) to enable asymptotic worst-case analysis of error propagation of BA-SC-CCs. We show that stronger component CCs exacerbate error propagation. Self-recovery threshold is general and can be used to investigate error propagation of other spatially coupled codes.
- Fourth, we introduce a threshold-based Max-Log-MAP (T-Max-Log-MAP) decoder for CCs. We then propose a decoder for BA-SC-CCs based on non-uniform quantization (NUQ), T-Max-Log-MAP decoder, and the TAMSA. The new decoder is referred to as the quantized T-Max-Log-MAP-TAMSA.

We present extensive numerical results to demonstrate the advantages of the BA-SC-CCs and the new decoder. Comparisons show that the terminated BA-SC-CCs perform as well as the sparsely braided convolutional code and perform better than the SC-LDPC codes, existing BMST codes, and the SC-RCC codes. In particular, for a decoding latency of 6,000 bits, the BA-SC-CC performs 0.5 dB better than the (3,6)-regular SC-LDPC code. The numerical results also show that, for low-resolution quantization, performance gains about 0.2 dB can be obtained with the proposed quantized T-Max-Log-MAP-TAMSA and 5-bits non-uniform quantization is sufficient for the quantized T-Max-Log-MAP-TAMSA to approach the performance of the floating-point based decoder.

II. BLOCK-ORIENTED ACCUMULATOR BASED SPATIALLY COUPLED CONVOLUTIONAL CODES

In this section, we describe the encoding (as shown in Fig. 1) and decoding processes of block-oriented accumulator based spatially coupled convolutional codes (BA-SC-CCs).

A. Encoding of a BA-SC-CC

Consider a CC with k inputs and n outputs. Let v denote the *overall constraint length* of the CC [52]. In this paper, terminated convolutional codes are used. We use $C_{n,k}[N, K]$ to denote a *terminated* k -input, n -output convolutional code when K bits are encoded together. Here, N denotes the length of the encoded output corresponding to K information bits. Then $C_{n,k}[N, K]$ can be decoded with the BCJR decoder [53].

Algorithm 1 Encoding of a BA-SC-CC

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- 1) **Initialization:** For $t < 0$, set $\mathbf{c}^{(t)} = \mathbf{0} \in \mathbb{F}_2^N$.
 - 2) **Recursion:** For $t = 0, 1, \dots, L-1$,
 - *Encoding:* Encode $\mathbf{u}^{(t)}$ by the encoder of the CC $\mathcal{C}_{n,k}[N, K]$ to obtain $\mathbf{v}^{(t)} \in \mathbb{F}_2^N$.
 - *Interleaving and superposition:* Compute the t -th transmitted block $\mathbf{c}^{(t)}$ as

$$\mathbf{c}^{(t)} = \mathbf{v}^{(t)} + \Pi(\mathbf{c}^{(t-1)}), \quad (1)$$
 where $\Pi(\mathbf{c}^{(t-1)})$ is the interleaved version of $\mathbf{c}^{(t-1)}$ with the interleaver Π .
 - 3) **Termination:** For $t = L, L+1, \dots, L+T-1$, set $\mathbf{u}^{(t)} = \mathbf{0} \in \mathbb{F}_2^K$ and compute $\mathbf{c}^{(t)}$ following the Step. 2).
-

Let $\mathbf{u} = (\mathbf{u}^{(0)}, \mathbf{u}^{(1)}, \dots, \mathbf{u}^{(L-1)})$ be L blocks of data to be transmitted, where $\mathbf{u}^{(t)} \in \mathbb{F}_2^K$. Here, L is referred to as the coupling length. The encoding algorithm of a BA-SC-CC is described in Algorithm 1, where Π is an interleaver of size N . For termination, T length- K all-zero sequences are taken as the input of the encoder.

It can be seen that the rate of a BA-SC-CC is

$$R = \frac{LK}{(L+T)N} = \frac{1}{1+T/L} \frac{K}{N}, \quad (2)$$

which is slightly less than the rate of the terminated CC. However, similar to SC-LDPC codes and braided convolutional codes, this rate loss becomes vanishingly small as the *termination overhead* $T/L \rightarrow 0$. Note that the rates of the terminated convolutional codes approach $\frac{K}{N}$ as $K \rightarrow \infty$. Hence, the rate of a BA-SC-CC approaches $\frac{K}{N}$, the rate of the unterminated CC, when $T/L \rightarrow 0$ and $K \rightarrow \infty$. Due to the recursive characteristic of BA-SC-CC, the last T output sub-blocks are usually non-zero.

Remark 1: It can be seen that BA-SC-CCs differ from spatially coupled serially concatenated codes (SC-SCC) [10], braided convolutional codes (BCC) [11], and braided block codes (BBC) [54] in coupling approach. In SC-SCC, the coupling is achieved by taking a portion of the output of the *outer encoder* at time t as the input of the *inner encoder* at time $t+1$ [10]. In BCC, turbo codes are coupled by feeding a portion of the output of the *upper encoder* at time t into the *lower encoder* at time $t+1$, and simultaneously feeding a portion of the output of the lower encoder at time t into the upper encoder at time $t+1$ [11]. BBC is typically treated as the coupled version of the product codes. In BBC, two constituent block codes are mutually interleaved, with the parity outputs of one code used as part of the input to the other in a *temporally shifted manner*, forming a braided encoding structure [54].

Remark 2: In BA-SC-CCs, a block-oriented accumulator is used to introduce coupling across multiple codewords of a terminated CC. This structure can be interpreted as a coupled extension of convolutional accumulate codes [55], where the bit-wise accumulator is replaced with a block-wise accumulator. It may also be viewed as a special case of BMST codes [13], in which the terminated CC acts as the component code. However, in this work, we do not frame BA-SC-CC within the BMST paradigm, as our objective is

to emphasize its unique role in coupling convolutional codes. In particular, the proposed block-oriented accumulation of a terminated convolutional code admits a structural interpretation as a periodic SC-LDPC code. Specifically, the terminated convolutional code first applies a linear transformation to the input information blocks, after which a memory-one, block-oriented accumulator creates a periodic syndrome-former. Terminated convolutional codes are considered because their trellis structure is essential for the free distance analysis, a property not shared by many alternative block codes. They also admit efficient MAP decoding, whereas optimal decoding for many powerful block codes, such as BCH or LDPC codes, is typically impractical. The proposed threshold-based Max-Log-MAP decoder in Sec. V is built directly upon this MAP decoding framework. We emphasize that different component codes may be preferable in different parameter regimes. The purpose of this work is not to claim universal superiority of BA-SC-CCs, but to rigorously analyze a class of spatially coupled convolutional codes. This serves as an important step toward a rigorous structural understanding of BMST codes.

B. Iterative Sliding Window Decoder

Let $\mathbf{y}^{(t)}$ denote the received signal sequence corresponding to $\mathbf{c}^{(t)}$. Given $\mathbf{y} = (\mathbf{y}^{(0)}, \mathbf{y}^{(1)}, \dots, \mathbf{y}^{(L+T-1)})$, the receiver attempts to recover $\mathbf{u}$. Similar to the decoding of SC-LDPC codes, this recovery process can be achieved with a sliding window decoder over the normal graph of the BA-SC-CCs. We present in Fig. 2 the normal graphical realization of a BA-SC-CC. This normal graph consists of four types of constraint nodes, which are node CC, node $=$, node Π , and node $+$. The normal graph is described in detail in the following.

- Node $=$, which has degree *three*, represents the constraint that all connecting variables must take the same value. Nodes $=$ are referred to as *variable nodes*.
- Node $+$, which has degree *three*, represents the constraint of the superposition relationship among the connecting variables. Nodes $+$ are referred to as *check nodes*.
- Node Π , which has degree *two*, interleaves or de-interleaves the input messages.
- Node CC, which has degree *two*, represents the constraint specified by the CC.
- The normal graph can be divided into layers, where each layer typically consists of a CC node, a check node, a Π node, and a variable node.

Let d denote the decoding delay and W denote the decoding window size. For a decoding delay of d , the sliding window decoder works iteratively over a subgraph consisting of $W = d+1$ consecutive layers of the normal graph. An example of the sliding window decoder with $d = 2$ and $W = 3$ is illustrated in Fig. 2. We call the first layer in a decoding window the *target layer*.

For the sliding window decoder, the messages are updated in a layer-by-layer manner. An iteration of the sliding window decoder consists of a forward iteration and a backward iteration with the schedule shown in Fig. 3. Here, the t -th layer is the target layer. Within a decoding layer, the messages

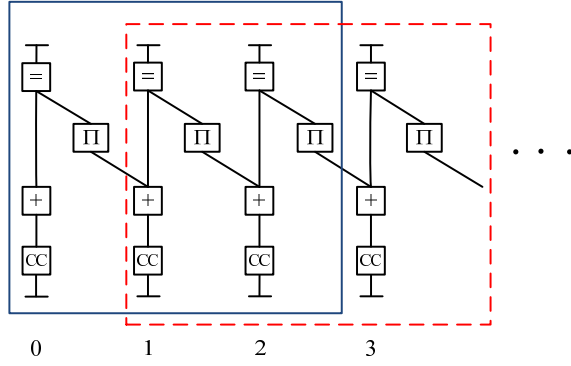


Fig. 2. Normal graph of a BA-SC-CC. The decoding window size is $W = 3$.

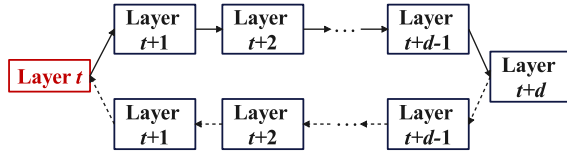


Fig. 3. The decoding schedule of the sliding window decoder in an iteration.

are updated according to the schedule $= \rightarrow + \rightarrow \text{CC} \rightarrow + \rightarrow =$. The node CC is decoded with the BCJR algorithm [53]. It can be seen that for a target layer the number of BCJR decoding per iteration is $2d$. Note that not all messages of constraint nodes need to be computed during decoding. To start the decoder, the messages over the half edges connecting to the nodes $=$ are initialized as the soft information from the received sequences and the messages over all the other edges are initialized as uniformly distributed binary variables. The decoding process proceeds until the maximum iteration number is reached or the *entropy-based stopping criterion* is satisfied. Let $h_{\ell-1}$ and h_{ℓ} denote the average entropies of the decoding outputs of the $(\ell-1)$ -th and the ℓ -th iterations, respectively. The entropy-based stopping criterion is satisfied when $|h_{\ell-1} - h_{\ell}|$ is less than a given threshold. A formal definition of entropy-based stopping criterion can be found in [13].

III. ANALYSIS OF BA-SC-CCs

For ease of presentation, we only consider 1-input, n -output terminated convolutional codes with memory order m in this section. For such codes, the overall constraint length v is equal to m [56] and termination can be achieved by appending an appropriately selected length- m bit sequence at the end of the information sequence. For feedforward CC, the appended termination sequence is the length- m all-zero sequence. For feedback CC, the appended termination sequence depends on the final state after encoding the K information bits. For convenience, we use $\mathcal{C}_t[(K+m)n, K, m]$ to denote the terminated convolutional code used in BA-SC-CCs, where m denotes the memory order of the CC. The unterminated convolutional code corresponding to $\mathcal{C}_t[(K+m)n, K, m]$ is denoted as $\mathcal{C}$. Note that our analytical results are applicable for any terminated convolutional codes since the analyses do not rely on their input numbers k . For convenience, we assume that L length- K sub-blocks are encoded jointly and T all-zero length- K sub-blocks are used for termination.

A. Threshold Analysis of BA-SC-CCs

For a fixed K , we obtain a BA-SC-CC ensemble by selecting the interleaver in each transmission uniformly at random. We call this ensemble the $(\mathcal{C}, K, L, T)$ -BA-SC-CC ensemble. By increasing the information length K , we can construct a sequence of $(\mathcal{C}, K, L, T)$ -BA-SC-CC ensembles. To analyze the asymptotic performance of $(\mathcal{C}, K, L, T)$ -BA-SC-CC ensembles when $K \rightarrow \infty$, we use the extrinsic information transfer (EXIT) chart.

Consider binary phase-shift keying (BPSK) modulation and additive white Gaussian noise (AWGN) channels. Without loss of generality, we assume that the messages are represented in the form of log-likelihood ratios (LLRs). For notation, we use I_{AP} to denote the MI between the *a posteriori* LLR and the corresponding codeword bit, I_A to denote the MI between the *a priori* LLR and the codeword bit, and I_E to denote the MI between the *extrinsic* LLR and the codeword bit. Following [57], the *a posteriori* LLR is assumed to be Gaussian distributed. To carry out EXIT chart analysis, we need to compute the transfer function of BCJR decoder of the terminated CC. In this paper, we adopt a simulation-based approach to obtain the transfer function, as no closed-form expression is available [58]. Assume that the length- K sequence $\mathbf{u}$ is encoded into a length- N sequence $\mathbf{v}$ by the terminated CC. Let the vector $\mathbf{L}_A$ denote *a priori* LLRs of the coded sequence $\mathbf{v}$. After executing the BCJR decoder, the output extrinsic LLRs are denoted by $\mathbf{L}_E$. Then the extrinsic MI can be approximated by the time average

$$I_E \approx 1 - \frac{1}{N} \sum_{i=1}^N \log_2 \left(1 + e^{(2v_i - 1) \cdot L_E(i)} \right),$$

where $L_E(i)$ denotes the i -th component of $\mathbf{L}_E$. The I_{AP} of the BCJR decoder of the terminated CC can be estimated similarly.

For a given E_b/N_o , the LLR obtained from the channel output is Gaussian distributed with the variance $\sigma_{\text{ch}}^2 = 8R \frac{E_b}{N_o} = 4\text{SNR}$ [57], where SNR denotes the signal-to-noise ratio (SNR). The corresponding MI is given as $I_{\text{ch}} = J(\sigma_{\text{ch}})$, where $J(\cdot)$ is defined in [57]. For a given E_b/N_o (or SNR), we can use the following algorithm to compute the asymptotic iterative decoding threshold of the $(\mathcal{C}, K, L, T)$ -BA-SC-CC ensemble for a given decoding delay d .

The asymptotic threshold $(E_b/N_o)^*$ of the BA-SC-CC ensembles is defined as the minimum E_b/N_o at which **Algorithm 2** declares a success. The corresponding threshold in terms of SNR is denoted as SNR^* .

B. Free Distance of a BA-SC-CC

It can be seen from Fig. 2 that the encoding output $\mathbf{c} = (\mathbf{c}^{(0)}, \mathbf{c}^{(1)}, \dots, \mathbf{c}^{(L+T-1)})$ satisfies the local constraints

$$(\Pi(\mathbf{c}^{(t-1)}) + \mathbf{c}^{(t)})\mathbf{H}^T = \mathbf{0}, \quad t = 0, 1, \dots, L+T-1,$$

where $\mathbf{H}$ is the parity-check matrix of the terminated convolutional code $\mathcal{C}_{n,k}[N, K]$. In the following, we prove that the free distance $d_{\min}$ of a $(\mathcal{C}, K, L, T)$ -BA-SC-CC is larger than the free distance d_{free} of the underlying CC $\mathcal{C}$ when certain condition is satisfied.

Algorithm 2 EXIT Analysis of a BA-SC-CC Ensemble

- 1) **Initialization:** All messages over the half-edges connected to nodes = are initialized as I_{ch} and all messages over the remaining edges, including half-edges connected to nodes CC and inter-connected full-edges, are initialized as 0. Select a maximum number of iterations I_{max} and a small positive number δ .
- 2) **Recursion:** For $t = 0, 1, \dots, L - 1$,
 - For the t -th decoding window, perform MI message processing/passing layer-by-layer with I_{max} iterations according to the schedule

$$= \rightarrow + \rightarrow CC \rightarrow + \rightarrow =.$$
 - If $(1.0 - I_{AP}) \geq \delta$, exit the iterations.
- 3) **Declaration:** If $(1.0 - I_{AP}) < \delta$, declare a success; else, declare a failure.

Noting that there is a one-to-one correspondence between the set of trellis paths and the set of codewords of $\mathcal{C}$. For a non-zero codeword $\mathbf{v} \in \mathcal{C}$, we use $w_H(\mathbf{v})$ to denote the Hamming weight of $\mathbf{v}$ and $p(\mathbf{v})$ to denote the number of edges in its path that are different from the all-zero path. The following lemma from [56] gives an upper bound on the path length of the all-zero sub-path nowhere merged with the zero state.

Lemma 1 ([56]): For the minimal encoder generating $\mathcal{C}$, the longest all-zero sub-path nowhere merged with the zero state has length upper bounded by $v - 1$, where v denotes the overall constraint length of the minimal encoder.

Accordingly, we have the following proposition.

Proposition 1: For any non-zero codeword $\mathbf{v} \in \mathcal{C}$, we have

$$w_H(\mathbf{v}) \leq p(\mathbf{v}) \leq w_H(\mathbf{v})v + 1,$$

where v denotes the overall constraint length of a minimal encoder generating $\mathcal{C}$.

Proof: The left part of the inequality is obvious. The upper bound of $p(\mathbf{v})$ is achieved when each edge of $\mathbf{v}$ with a nonzero output is followed by an all-zero path of length $v - 1$. Since there are at most $w_H(\mathbf{v})$ edges with nonzero outputs in the trellis path of $\mathbf{v}$, we obtain this upper bound. ■

According to Proposition 1, it can be easily shown that for $\mathbf{v} \in \mathcal{C}$ with $w_H(\mathbf{v}) = d_{\text{free}}$, we have

$$d_{\text{free}} \leq p(\mathbf{v}) \leq d_{\text{free}}v + 1.$$

For convenience, we use $\text{div}(\mathbf{v})$ to denote the number of times the path of $\mathbf{v}$ that diverges from and remerges with the all-zero path. Obviously, we have $\text{div}(\mathbf{0}) = 0$ and $\text{div}(\mathbf{v}) = 1$ when $w_H(\mathbf{v}) = d_{\text{free}}$. We have the following lemma.

Lemma 2: If $\text{div}(\mathbf{v}) \neq 0$, then $w_H(\mathbf{v}) \geq \text{div}(\mathbf{v})d_{\text{free}}$.

Proof: When $\text{div}(\mathbf{v}) \neq 0$, $\mathbf{v}$ can be obtained as the summation of $\text{div}(\mathbf{v})$ nonzero codewords whose nonzero spans are disjoint. As a result, we have

$$w_H(\mathbf{v}) \geq \text{div}(\mathbf{v})d_{\text{free}}.$$

■

According to Lemma 2, we have $\text{div}(\mathbf{v}) = 1$ when $0 < w_H(\mathbf{v}) < 2d_{\text{free}}$. Theorem 3 gives a lower bound on the free distance d_{min} of a $(\mathcal{C}, K, L, T)$ -BA-SC-CC.

Definition 1: The interleaver Π of length N is a doubly dispersed interleaver if $\forall i, j \in \{0, 1, \dots, N - 1\}$ with $i \neq j$ and $|i - j| < S_0$, the conditions

$$|\Pi(i) - \Pi(j)| \geq S_1 \quad (3)$$

$$|\Pi(\Pi(i)) - \Pi(\Pi(j))| \geq S_2 \quad (4)$$

are satisfied, where $S_0 \leq S_1 \leq S_2$.

Theorem 3: The free distance d_{min} of the $(\mathcal{C}, K, L, T)$ -BA-SC-CC satisfies

$$d_{\text{min}} \geq 3d_{\text{free}} - 2$$

when Π is a doubly dispersed interleaver with $S_0 \geq 3vd_{\text{free}}n$.

Proof: Without loss generality, let $\mathbf{c}^{(t)}$ be the first non-zero sub-vector of $\mathbf{c}$. That is $\mathbf{c}^{(i)} = \mathbf{0}$ for $i = 0, 1, \dots, t - 1$. According to the encoding procedure, we have

$$\begin{aligned} \mathbf{c}^{(t)} &= \mathbf{v}^{(t)}, \\ \mathbf{c}^{(t+1)} &= \mathbf{v}^{(t+1)} + \Pi(\mathbf{v}^{(t)}), \\ \mathbf{c}^{(t+2)} &= \mathbf{v}^{(t+2)} + \Pi(\mathbf{v}^{(t+1)}) + \Pi^2(\mathbf{v}^{(t)}). \end{aligned}$$

Case 1: $w_H(\mathbf{c}^{(t)}) \geq 3d_{\text{free}}$.

In this case, we have

$$w_H(\mathbf{c}) \geq w_H(\mathbf{c}^{(t)}) \geq 3d_{\text{free}}. \quad (5)$$

Case 2: $2d_{\text{free}} \leq w_H(\mathbf{c}^{(t)}) < 3d_{\text{free}}$.

In this case, we have $1 \leq \text{div}(\mathbf{v}^{(t)}) < 3$. That is, $\text{div}(\mathbf{v}^{(t)}) = 1$ or $\text{div}(\mathbf{v}^{(t)}) = 2$. When $\text{div}(\mathbf{v}^{(t)}) = 2$, $\mathbf{v}^{(t)}$ can be written as

$$\mathbf{v}^{(t)} = \tilde{\mathbf{v}}^{(t)} + \hat{\mathbf{v}}^{(t)},$$

where $\tilde{\mathbf{v}}^{(t)}$ and $\hat{\mathbf{v}}^{(t)}$ are codewords of the terminated convolutional code $\mathcal{C}$ and $\text{div}(\tilde{\mathbf{v}}^{(t)}) = \text{div}(\hat{\mathbf{v}}^{(t)}) = 1$. Furthermore, according to Proposition 1, we have $2d_{\text{free}} \leq p(\mathbf{v}^{(t)}) \leq 3d_{\text{free}}v$.

Case 2.1: $w_H(\mathbf{v}^{(t+1)}) = 0$.

According to the encoding procedure, we have $\mathbf{c}^{(t+1)} = \Pi(\mathbf{v}^{(t)})$, which means that

$$w_H(\mathbf{c}) \geq 2w_H(\mathbf{v}^{(t)}) \geq 4d_{\text{free}}. \quad (6)$$

Case 2.2: $w_H(\mathbf{v}^{(t+1)}) > 0$.

We consider a sub-path of length $d_{\text{free}}v$ which starts from the first nonzero edge of $\mathbf{v}^{(t+1)}$. According to Lemma 1, the Hamming weight of the output subsequence of this sub-path is at least d_{free} . Let assume that $3d_{\text{free}}vn \leq S_0 \leq S_1$. When $\text{div}(\mathbf{v}^{(t)}) = 1$, the number of ones of $\mathbf{v}^{(t+1)}$ within the considered interval that are cancelled out by adding $\Pi(\mathbf{v}^{(t)})$ is at most 1. When $\text{div}(\mathbf{v}^{(t)}) = 2$, the number of ones of $\mathbf{v}^{(t+1)}$ within the considered interval that are cancelled out by adding $\Pi(\mathbf{v}^{(t)})$ is at most 2. Hence, we have

$$w_H(\mathbf{c}^{(t+1)}) \geq d_{\text{free}} - 2.$$

That is,

$$w_H(\mathbf{c}) \geq w_H(\mathbf{c}^{(t)}) + w_H(\mathbf{c}^{(t+1)}) \geq 3d_{\text{free}} - 2. \quad (7)$$

Based on (6) and (7), we have

$$w_H(\mathbf{c}) \geq 3d_{\text{free}} - 2, \quad (8)$$

when $2d_{\text{free}} \leq w_H(\mathbf{c}^{(t)}) < 3d_{\text{free}}$ and $3d_{\text{free}}vn \leq S_0 \leq S_1$.

Case 3: $d_{\text{free}} \leq w_H(\mathbf{c}^{(t)}) < 2d_{\text{free}}$.

In this case, we have $\text{div}(\mathbf{c}^{(t)}) = 1$. This condition indicates that $p(\mathbf{c}^{(t)}) \leq 2d_{\text{free}}v$, which means that the length of the nonzero span of $\mathbf{c}^{(t)}$ is at most

$$p(\mathbf{v}^{(t)})n \leq 2d_{\text{free}}vn. \quad (9)$$

Case 3.1: $d_{\text{free}} \leq w_H(\mathbf{v}^{(t+1)}) < 2d_{\text{free}}$.

According to Lemma 2, we have $\text{div}(\mathbf{v}^{(t+1)}) = 1$. Similar to the derivation of (9), it can be shown that the length of the nonzero span of $\mathbf{v}^{(t+1)}$ is at most

$$p(\mathbf{v}^{(t+1)})n \leq 2d_{\text{free}}vn.$$

When Π is a doubly dispersed interleaver with $S_1 \geq S_0 \geq 2d_{\text{free}}vn$, the number of ones of $\mathbf{v}^{(t+1)}$ that are canceled out in the summation $\mathbf{v}^{(t+1)} + \Pi(\mathbf{v}^{(t)})$ is at most 1. As a result, we have

$$w_H(\mathbf{c}^{(t+1)}) \geq 2d_{\text{free}} - 2.$$

That is, we have

$$w_H(\mathbf{c}) \geq w_H(\mathbf{c}^{(t)}) + w_H(\mathbf{c}^{(t+1)}) \geq 3d_{\text{free}} - 2. \quad (10)$$

Case 3.2: $w_H(\mathbf{v}^{(t+1)}) \geq 2d_{\text{free}}$.

When $\text{div}(\mathbf{v}^{(t+1)}) = 1$, we consider a sub-path of length $2d_{\text{free}}v$ which starts from the first nonzero edge of $\mathbf{v}^{(t+1)}$. According to Lemma 1, the Hamming weight of the output subsequence of this sub-path is at least $2d_{\text{free}}$. When Π is a doubly dispersed interleaver with $S_1 \geq S_0 \geq 2d_{\text{free}}vn$, the number of ones of $\mathbf{v}^{(t+1)}$ in the sub-sequence corresponding to the considered sub-path that are cancelled out in the summation $\mathbf{v}^{(t+1)} + \Pi(\mathbf{v}^{(t)})$ is at most 1. This indicates that

$$w_H(\mathbf{c}^{(t+1)}) \geq 2d_{\text{free}} - 1,$$

which means that

$$w_H(\mathbf{c}) \geq w_H(\mathbf{c}^{(t)}) + w_H(\mathbf{c}^{(t+1)}) \geq 3d_{\text{free}} - 1. \quad (11)$$

When $\text{div}(\mathbf{v}^{(t+1)}) \geq 2$, we use $\tilde{\mathbf{v}}^{(t+1)}$ to denote the codeword corresponding to the first divergence of the path of $\mathbf{v}^{(t+1)}$ with the all-zero path and $\hat{\mathbf{v}}^{(t+1)}$ to denote the codeword corresponding to the second divergence of the path of $\mathbf{v}^{(t+1)}$ with the all-zero path. Hence, we have

$$\text{div}(\tilde{\mathbf{v}}^{(t+1)}) = \text{div}(\hat{\mathbf{v}}^{(t+1)}) = 1.$$

Similar to the derivation of the case with $\text{div}(\mathbf{v}^{(t+1)}) = 1$, we consider a sub-path of length $d_{\text{free}}v$ which starts from the first nonzero edge of $\tilde{\mathbf{v}}^{(t+1)}$. According to Lemma 1, the Hamming weight of the output subsequence of this sub-path is at least d_{free} . When Π is a doubly dispersed interleaver with $S_1 \geq S_0 \geq d_{\text{free}}vn$, the number of ones of $\tilde{\mathbf{v}}^{(t+1)}$ that are canceled out in the summation $\tilde{\mathbf{v}}^{(t+1)} + \Pi(\mathbf{v}^{(t)})$ is at most 1. Similarly, it can be shown that the number of ones of $\hat{\mathbf{v}}^{(t+1)}$ that are canceled out in the summation $\hat{\mathbf{v}}^{(t+1)} + \Pi(\mathbf{v}^{(t)})$ is at most 1. Hence, we have $w_H(\mathbf{c}^{(t+1)}) \geq 2d_{\text{free}} - 2$, which means that

$$w_H(\mathbf{c}) \geq w_H(\mathbf{c}^{(t)}) + w_H(\mathbf{c}^{(t+1)}) \geq 3d_{\text{free}} - 2. \quad (12)$$

Case 3.3: $w_H(\mathbf{v}^{(t+1)}) = 0$.

We have $\mathbf{c}^{(t+1)} = \Pi(\mathbf{c}^{(t)})$. When $\mathbf{v}^{(t+2)} = \mathbf{0}$, we have

$$\mathbf{c}^{(t+2)} = \Pi(\Pi(\mathbf{c}^{(t)})),$$

which indicates that

$$w_H(\mathbf{c}) \geq w_H(\mathbf{c}^{(t)}) + w_H(\mathbf{c}^{(t+1)}) + w_H(\mathbf{c}^{(t+2)}) \geq 3d_{\text{free}}. \quad (13)$$

When $\mathbf{v}^{(t+2)} \neq \mathbf{0}$, we consider the codeword $\tilde{\mathbf{v}}^{(t+2)}$ corresponding to the first divergence of the path of $\mathbf{v}^{(t+2)}$ with the all-zero path. Then we have

$$\text{div}(\tilde{\mathbf{v}}^{(t+2)}) = 1.$$

For ease of presentation, we write $\mathbf{v}^{(t+2)}$ as $\mathbf{v}^{(t+2)} = \tilde{\mathbf{v}}^{(t+2)} + \hat{\mathbf{v}}^{(t+2)}$. Then we have

$$\begin{aligned} \mathbf{c}^{(t+2)} &= \mathbf{v}^{(t+2)} + \Pi(\Pi(\mathbf{v}^{(t)})) = \tilde{\mathbf{v}}^{(t+2)} + \hat{\mathbf{v}}^{(t+2)} + \Pi(\Pi(\mathbf{v}^{(t)})). \end{aligned}$$

Similarly, we consider a sub-path of length $d_{\text{free}}v$ which starts from the first nonzero edge of $\tilde{\mathbf{v}}^{(t+2)}$. According to Lemma 1, the Hamming weight of the output subsequence of this sub-path is at least d_{free} . Hence, when Π is a doubly dispersed interleaver with $S_2 \geq S_1 \geq S_0 \geq d_{\text{free}}vn$, the number of ones in $\tilde{\mathbf{v}}^{(t+2)}$ that are cancelled out in the summation $\tilde{\mathbf{v}}^{(t+2)} + \hat{\mathbf{v}}^{(t+2)} + \Pi(\Pi(\mathbf{v}^{(t)}))$ is at most 1. This result indicates that

$$w_H(\mathbf{c}) \geq 3d_{\text{free}} - 1. \quad (14)$$

Inequalities (10), (11), (12), (13), and (14) indicate that

$$w_H(\mathbf{c}) \geq 3d_{\text{free}} - 2 \quad (15)$$

when $d_{\text{free}} \leq w_H(\mathbf{c}^{(t)}) < 2d_{\text{free}}$ and $2d_{\text{free}}vn \leq S_0 \leq S_1 \leq S_2$.

Inequalities (5), (8), and (15) conclude the proof of this theorem. ■

We give a construction of doubly dispersed interleavers in the following. Let Π be the interleaver defined by $\Pi(i) = pi \bmod N$, where p and N are coprime and $p < \sqrt[3]{N}$. When $|i - j| < p$, we have

$$|\Pi(i) - \Pi(j)| = |p(i - j)| \bmod N,$$

which means that $p \leq |\Pi(i) - \Pi(j)| < p^2$. Furthermore, we have

$$|\Pi(\Pi(i)) - \Pi(\Pi(j))| = |p^2(i - j)| \bmod N,$$

which means that $p^2 \leq |\Pi(\Pi(i)) - \Pi(\Pi(j))| < p^3$. For this interleaver, we have $S_0 = p$. It can be seen that $\Pi(i) = pi \bmod N$ is a doubly dispersed interleaver with length larger than p^3 , which however is too long since $S_0 = p > 2d_{\text{free}}vn$. It is interesting to study the minimum length N required to construct doubly dispersed interleavers.

Consider the rate-half recursive systematic convolutional code $\mathcal{C}_1$ specified by the generator matrix

$$G_1(D) = \begin{bmatrix} 1 & \frac{1+D^2}{1+D+D^2} \end{bmatrix}. \quad (16)$$

We have $n = 2$, $m = v = 2$, $d_{\text{free}} = 5$. Then, the free distance of the resulting BA-SC-CC is lower bounded by 13. Furthermore, the condition in Theorem 3 is $S_0 > 60$. However,

TABLE I
THE MAPPING OF THE INTERLEAVER OF LENGTH 34

i	1	2	3	4	5	6	7	8	9
$\Pi(i)$	15	25	33	14	24	10	30	8	34
i	10	11	12	13	14	15	16	17	18
$\Pi(i)$	6	4	17	19	11	7	3	1	31
i	19	20	21	22	23	24	25	26	27
$\Pi(i)$	22	12	23	9	29	27	32	20	5
i	28	29	30	31	32	33	34		
$\Pi(i)$	16	13	2	28	18	21	26		

TABLE II
THE MAPPING OF THE INTERLEAVER OF LENGTH 44

i	1	2	3	4	5	6	7	8	9
$\Pi(i)$	15	25	40	33	14	24	10	30	8
i	10	11	12	13	14	15	16	17	18
$\Pi(i)$	37	34	6	41	4	43	17	36	19
i	19	20	21	22	23	24	25	26	27
$\Pi(i)$	11	7	38	3	1	31	22	12	23
i	28	29	30	31	32	33	34	35	36
$\Pi(i)$	9	29	27	32	20	5	35	16	39
i	37	38	39	40	41	42	43	44	
$\Pi(i)$	44	13	2	28	42	18	21	26	

TABLE III
THE FREE DISTANCES OF EIGHT BA-SC-CCs

K	L	T	$d_{\min}$
15	4	2	14
15	4	3	17
15	6	2	14
15	6	3	17
20	4	2	15
20	4	3	18
20	6	2	15
20	6	3	18

the doubly dispersed interleaver with $S_0 > 60$ is too large to allow exact computation of the free distance. Therefore, we examine several smaller examples to partially demonstrate the tightness of the proposed lower bound. Specifically, we consider $K = 15, 20$, $L = 4, 6$, and $T = 2, 3$. For $K = 15$, the interleaver Π of length 34 is defined in Table I. For $K = 20$, the interleaver of length 44 is defined in Table II. The free distances of these eight BA-SC-CCs based on $\mathcal{C}_1$ are computed using the algorithm in [59], and the results are reported in Table III. As shown, the free distances in all considered cases

are at least 14, which is slightly higher than the lower bound of 13. In addition, the results indicate that the free distance depends on the parameter T .

Remark 3: It is well-known that analyzing the weight distributions of a given spatially coupled code is challenging [3], [47]. Existing results indicate that spatial coupling does not decrease the minimum distance of the uncoupled code [47]. In Theorem 3, we demonstrate that, by selecting the parameters appropriately, the free distance of a BA-SC-CC exceeds that of the underlying terminated convolutional code. Note that the property of convolutional codes in Lemma 1 is essential to the proof of Theorem 3. Hence, it cannot be directly extended to arbitrary linear block codes. The proposed analysis aims to characterize the structural impact of block-oriented accumulation. We emphasize that an increase in $d_{\min}$ for BA-SC-CCs does not, by itself, establish their superiority over other spatially coupled code families, such as BMST codes or SC-LDPC codes.

Remark 4: In Theorem 3, doubly dispersed interleavers are required. The use of two dispersions in this type of interleavers results in a large interleaving length, which is undesired in practice. Extending the proof to more general interleavers remains an important direction for future work.

C. Asymptotic Analysis of Error Propagation

Similar to braided convolutional codes, the BA-SC-CCs may experience decoder error propagation. That is, the decoding error of one layer may cause the decoding failures of its subsequent layers. The error propagation is rooted in the fact the current output sub-block depends on the previous output sub-blocks. In this subsection, we intend to analyze the influence of error propagation in long length regime.

Let the ρ -th layer be the first layer decoded in error. Since layers 0 through $\rho - 1$ are decoded correctly, they can be removed from the normal graph, resulting in the ρ -th layer having the same graphical structure as the 0-th layer. Therefore, without loss of generality, we assume $\rho = 0$. Due to block-oriented accumulation, the decoding performances of its subsequent layers will be affected. We are interested in the question that is it possible for the iterative sliding window decoder to successfully recover the t -th layer for some $t > 0$. Once the t -th layer is recovered, the harmful influence of the failed decoding of the 0-th layer is fully eliminated. To be specific, we analyze the minimum required SNR for the iterative sliding window decoder to recover from decoding failure. This minimum required SNR is referred to as the *self-recovery threshold* of a BA-SC-CC ensemble.

In this paper, the worst-case analysis of the self-recovery threshold is carried out. Specifically, we consider the self-recovery threshold of a $(\mathcal{C}, K, L, T)$ -BA-SC-CC ensemble under the assumption that the 0-th layer is erased. For simplicity, we restrict the decoder for recovery to be the iterative sliding window decoder, as opposed to the maximum likelihood decoder. Hence, this self-recovery threshold is referred to as *constrained self-recovery threshold* (CSRT). In practice, it is preferred that a decoding failure only affects the recovery of a finite number of layers. To be more specific, we consider the constrained self-recovery threshold with a window size

D , which is defined as the minimum required SNR at which the iterative sliding window decoder is capable of recovering a layer of the D layers with indices $1, 2, \dots, D$, where the assumption that the 0-th layer is erased is made. For convenience, the constrained self-recovery threshold with a window size D is denoted as $S_r(D)$. For convenience, we refer to D as the *maximum allowed recovery window size*.

The EXIT chart can be used to analyze the asymptotic behaviours of $S_r(D)$ of the $(\mathcal{C}, K, L, T)$ -BA-SC-CC ensemble when $K \rightarrow \infty$. The **Algorithm 2** can be modified to compute $S_r(D)$. The main modifications are summarized in the following.

- 1) The extrinsic mutual information transferred from the 0-th layer to the 1st layer is set to be zero, which corresponds to the assumption that the 0-th layer is erased.
- 2) The step **Recursion** of **Algorithm 2** is executed from $t = 1$ to $t = D$, which corresponds to the requirement that the maximum allowed recovery window size is D .

The asymptotic constrained self-recovery threshold $S_r(D)$ of the $(\mathcal{C}, +\infty, L, T)$ -BA-SC-CC ensemble is the minimum SNR at which the modified EXIT chart algorithm declares a success.

We introduce the self-recovery threshold to study the error propagation of the BA-SC-CCs. We believe that this concept may also be useful for studying error propagation of other classes of spatially coupled codes. For BA-SC-CCs, the self-recovery thresholds are obtained by the EXIT analysis in **Algorithm 2**.

IV. NUMERICAL RESULTS AND PERFORMANCE COMPARISONS

In this section, we present analytical and simulation results to illustrate the advantages of BA-SC-CCs. All results are obtained under the assumptions of BPSK modulation and AWGN channels. In the computation of the asymptotic iterative decoding thresholds, we set a maximum number of iterations $I_{\max} = 1,000$ per window to ensure the convergence of the EXIT chart analysis [58]. The parameter δ is set to 0.000001. In performance evaluations, the interleaver is randomly generated but fixed from frame to frame. The iterative sliding window decoder is performed using a layer-by-layer updating schedule with a maximum iteration number of 18, and the entropy-based stopping criterion with a threshold of 10^{-6} . Since BPSK modulation is assumed, we comparing the error correcting performances, we follow [60], [61] to define the decoding latency as the number of required received symbols to initiate decoding. For the sliding window decoder with a decoding delay of d , the decoding latency is $(d+1)(K+m)n$.

A. Iterative Decoding Thresholds

Consider the rate-half recursive systematic convolutional code $\mathcal{C}_1$ specified by (16). The iterative decoding thresholds of the $(\mathcal{C}_1, \infty, 595, T)$ -BA-SC-CC ensemble under different decoding delay d 's are given in Fig. 4. For a given decoding delay d , the termination length is set as $T = d$. It can be seen from Fig. 4 that

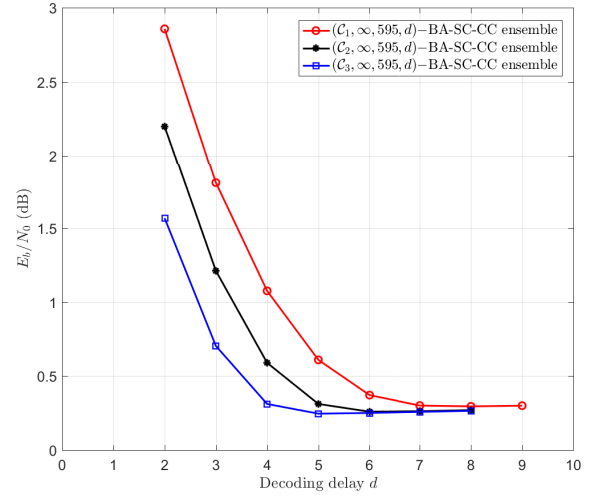


Fig. 4. The required decoding thresholds $(E_b/N_0)^*$ of different BA-SC-CC ensembles. The coupling lengths of all BA-SC-CC ensembles are $L = 595$.

- the threshold of the $(\mathcal{C}_1, \infty, 595, T)$ -BA-SC-CC ensemble improves as the decoding delay d increases and the improvement saturates for $d \geq 7$.
- the threshold of the $(\mathcal{C}_1, \infty, 595, 7)$ -BA-SC-CC ensemble is only 0.2 dB away from the channel capacity.

It is expected that we can further improve the thresholds by employing more powerful convolutional codes. To verify this, we show in Fig. 4 the thresholds of the BA-SC-CC ensemble based on convolutional code $\mathcal{C}_2$ specified by the generator matrix

$$G_2(D) = \left[1 \quad \frac{1+D+D^2+D^3}{1+D+D^3} \right],$$

and the thresholds of the BA-SC-CC ensemble based on convolutional code $\mathcal{C}_3$ specified by the generator matrix

$$G_3(D) = \left[1 \quad \frac{1+D+D^3+D^4}{1+D+D^2+D^3+D^4} \right].$$

It can be seen that, for a given decoding delay $d (< 9)$, the asymptotic threshold of the $(\mathcal{C}_2, \infty, 595, T)$ -BA-SC-CC ensemble is better than that of the $(\mathcal{C}_1, \infty, 595, T)$ -BA-SC-CC ensemble and the asymptotic threshold of the $(\mathcal{C}_3, \infty, 595, T)$ -BA-SC-CC ensemble is better than that of the $(\mathcal{C}_2, \infty, 595, T)$ -BA-SC-CC ensemble. Also note that the threshold of the $(\mathcal{C}_3, \infty, 595, 6)$ -BA-SC-CC ensemble is less than 0.1 dB away from the channel capacity. The negligible degradation in threshold observed for large values of d in Fig. 4 is due to the increased rate loss that accompanies larger d .

B. The Constrained Self-Recovery Thresholds

In this subsection, we investigate the impacts of the underlying CCs and the maximum allowed recovery window size D on the CSRT $S_r(D)$. We consider the BA-SC-CC ensembles built upon $\mathcal{C}_1$, $\mathcal{C}_2$, and $\mathcal{C}_3$. We present in Fig. 5 the CSRT $S_r(D)$ of different BA-SC-CC ensembles. The CSRT is computed by EXIT chart analysis. It can be observed from Fig. 5 that the better the underlying CC, the worse the CSRT $S_r(D)$. We also observe that for a given BA-SC-CC ensemble the

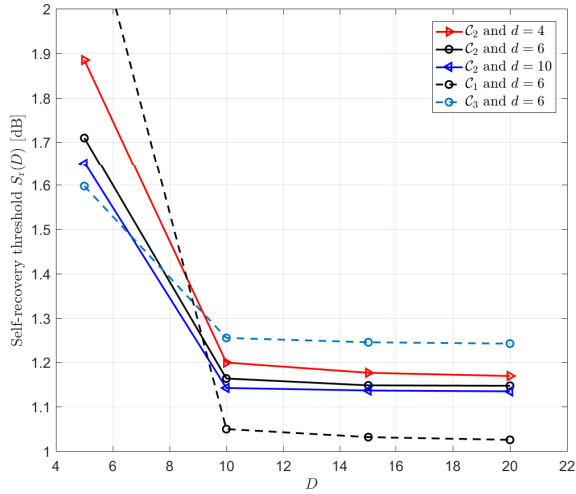


Fig. 5. The constrained self-recovery thresholds of different BA-SC-CC ensembles.

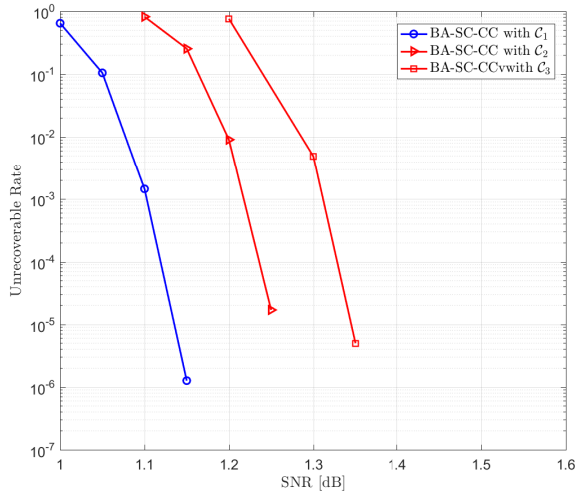


Fig. 6. The unrecoverable rates of different BA-SC-CCs for $D = 20$ and $W = 7$.

CSRT $S_r(D)$ improves with the increase of D . These results indicate that, for BA-SC-CCs, there is a trade-off between the iterative decoding threshold and the capability of combating error propagation.

To verify the correctness of the above analytical results, we have simulated the unrecoverable rate (UR) of the BA-SC-CC under the sliding window decoder. The AWGN channel and the BPSK modulation are assumed. For a frame of the BA-SC-CC, we execute the sliding window decoder to decode the layers with indices $1, 2, \dots, D$ under the assumption that the 0-th layer is erased. If a decoding success is declared for a layer, the error of the 0-th layer is recoverable; otherwise, the error of the 0-th layer is unrecoverable. The UR is defined as the ratio between the number of unrecoverable frames and the total number of simulated frames. That is,

$$\text{UR} = \frac{\text{Number of unrecoverable frames}}{\text{Total number of simulated frames}}.$$

We present in Fig. 6 the unrecoverable rate of different BA-SC-CCs, where we select $W = 7$, $D = 20$, and $K = 10,000$.

It can be seen that the BA-SC-CC based on C_1 performs the best in terms of UR and the BA-SC-CC based on C_2 performs better than the BA-SC-CC based on C_3 . These simulation results are consistent with the analytical results in Fig. 5.

C. Comparisons With SC-LDPC Codes

In this subsection, we compare the error performances of BA-SC-CCs with those of SC-LDPC codes. We simulate the performances of randomly selected BA-SC-CCs from the $(C_1, \infty, 595, 5)$ -BA-SC-CC ensemble and randomly selected BA-SC-CCs from the $(C_2, \infty, 645, 5)$ -BA-SC-CC ensemble. The considered decoding latencies include 6,000 bits, 15,000 bits, 20,000 bits, and 30,000 bits.

- For the decoding latency of 6,000 bits, the BA-SC-CCs based on the terminated convolutional codes $C_2[1006, 500, 3]$ and $C_1[1004, 500, 2]$ are simulated.
- For the decoding latency of 15,000 bits, the BA-SC-CCs based on the terminated convolutional codes $C_2[2506, 1250, 3]$ and $C_1[2504, 1250, 2]$ are simulated. The constraint lengths are about 5000.
- For the decoding latency of 20,000 bits, the BA-SC-CCs based on the terminated convolutional codes $C_2[3334, 1664, 3]$ and $C_1[3332, 1664, 2]$ are simulated. The constraint lengths are about 6664.
- For the decoding latency of 30,000 bits, the BA-SC-CCs based on the terminated convolutional codes $C_2[5006, 2500, 3]$ and $C_1[5004, 2500, 2]$ are simulated.

For the considered six BA-SC-CCs, we set $d = 5$ in the simulations.

The protograph-based SC-LDPC codes are selected for comparison [62]. We use M to denote the lifting factor. It is suggested in [63] and [64] that SC-LDPC codes with coupling memory being 1 and relatively small decoding delay have superior finite-length performance. Thus we consider $(3, 6)$ -regular SC-LDPC codes with $\mathbf{B}_0 = [2 \ 1]$ and $\mathbf{B}_1 = [1 \ 2]$, and $(4, 8)$ -regular SC-LDPC codes with $\mathbf{B}_0 = [3 \ 1]$ and $\mathbf{B}_1 = [1 \ 3]$. We do not consider the $(5, 10)$ -regular SC-LDPC codes since the decoding threshold improvement from $(4, 8)$ -regular SC-LDPC codes to $(5, 10)$ -regular SC-LDPC codes is small [62]. We set the decoding window sizes of $(3, 6)$ -regular SC-LDPC codes and $(4, 8)$ -regular SC-LDPC codes as $W = 6$ and $W = 4$, respectively, which are good choices when the decoding latencies are finite and fixed. For the above considered SC-LDPC codes, the decoding latencies and constraint lengths are given as $2MW$ and $2M$, respectively. For a given decoding latency and decoding window size, the lifting factor M is chosen accordingly. For example, to achieve a decoding latency of 20,000 bits, the lifting factors for $(3, 6)$ -regular SC-LDPC code and $(4, 8)$ -regular SC-LDPC code are $M = 1,666$ and $M = 2,500$, respectively.

The performances of the considered BA-SC-CCs and SC-LDPC codes are shown in Fig. 7, leading to the following observations.

- For decoding latencies of 6,000, 15,000 and 20,000 bits, BA-SC-CCs based on C_1 outperform those based on C_2 . Conversely, for latency 30,000 bits, BA-SC-CC based on

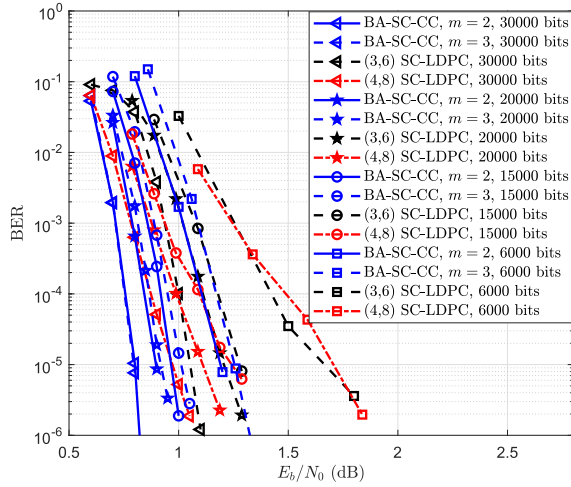


Fig. 7. Performance comparisons of BA-SC-CCs and SC-LDPC codes under the constraint of equal decoding latencies. These codes have comparable coding rates.

$\mathcal{C}_2$ achieves superior performance in the low error rate region.

- At the decoding latency of 6,000 bits, the BA-SC-CC based on $\mathcal{C}_1$ achieves a BER of 10^{-5} at 1.18 dB, offering gains of approximately 0.45 dB and 0.52 dB over the comparable (3,6)-regular and (4,8)-regular SC-LDPC codes, respectively.
- At the decoding latency of 15,000 bits, the BA-SC-CC based on $\mathcal{C}_1$ outperforms the comparable (3,6)-regular and (4,8)-regular SC-LDPC codes by about 0.30 dB and 0.27 dB, respectively.
- At the decoding latency of 20,000 bits, the BA-SC-CC based on $\mathcal{C}_1$ provides gains of about 0.35 dB and 0.30 dB over the comparable (3,6)-regular and (4,8)-regular SC-LDPC codes, respectively.
- At the decoding latency of 30,000 bits, the BA-SC-CC based on $\mathcal{C}_2$ surpasses the comparable (3,6)-regular and (4,8)-regular SC-LDPC codes by about 0.30 dB and 0.20 dB, respectively.

We also compare the performances under the constraint of comparable constraint lengths [5]. The considered constraint lengths include $v_s = 6664$ and $v_s = 5000$. For the (3,6)-regular and (4,8)-regular SC-LDPC codes, the lifting sizes are set as $M = 1666$ and $M = 1250$, respectively. The decoding window sizes are set to $W = 6$. The simulation results are given in Fig. 8.

For a constraint length of 6664, the BA-SC-CC based on $\mathcal{C}_1$ outperforms the comparable (3,6)-regular and (4,8)-regular SC-LDPC codes by approximately 0.32 dB and 0.33 dB, respectively. A similar performance advantage is observed at the constraint length of 5000, where the proposed BA-SC-CC based on $\mathcal{C}_1$ exceeds the corresponding SC-LDPC codes by about 0.33 dB and 0.43 dB, respectively.

These results demonstrate that BA-SC-CCs can offer superior performance to regular SC-LDPC codes across a broad range of latencies, with the optimal choice of component code depending on the target latency and performance regime.

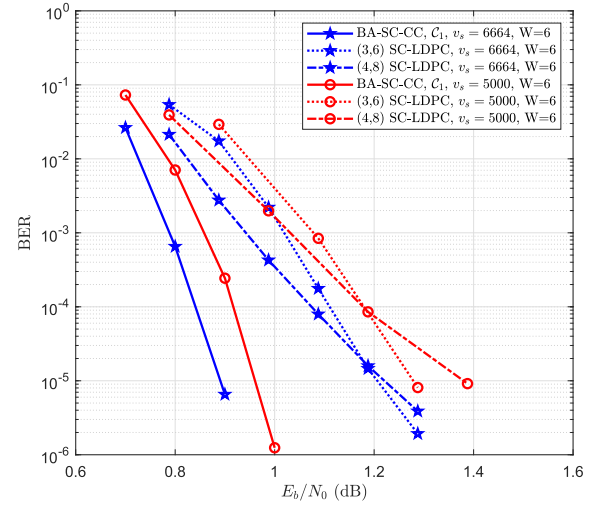


Fig. 8. Performance comparisons of BA-SC-CCs and SC-LDPC codes under the constraint of comparable constraint lengths. These codes have comparable coding rates.

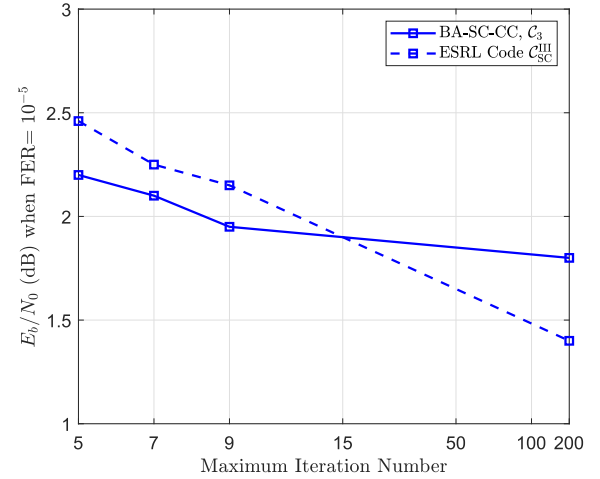


Fig. 9. Performance comparison between a BA-SC-CC and an ESRL code.

In the following, we compare the proposed scheme with a recently designed SC-LDPC code [23], namely the edge-spreading Raptor-like (ESRL) code. Specifically, we consider the ESRL code $\mathcal{C}_{SC}^{III}$, which has an information length of 85,800 and a coding rate of 0.52 [23]. For comparison, we select the BA-SC-CC constructed from $\mathcal{C}_3$ with parameters $K = 1,000$, $L = 84$, and $T = 2$, resulting in a code length and coding rate comparable to those of $\mathcal{C}_{SC}^{III}$. We adopt the simulation setup in [23] and perform decoding over the full normal graph with $L + T$ layers. The frame error rate (FER) is used as the performance metric. Fig. 9 shows the required E_b/N_0 to achieve an FER of 10^{-5} for different maximum iteration numbers. The results indicate that the BA-SC-CC slightly outperforms the ESRL code when the maximum number of iterations I_{max} is small, whereas the ESRL code performs better for larger I_{max} . We also note that, unlike ESRL codes, BA-SC-CCs do not require a dedicated design procedure.

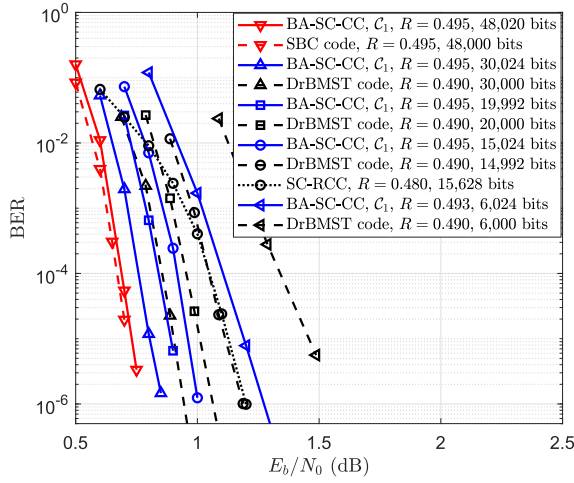


Fig. 10. Performance comparisons of BA-SC-CCs, SBC codes, BMST codes, and SC-RCC codes under the constraint of equal decoding latencies. These codes have comparable coding rates.

D. Comparisons With BMST Codes and SC-RCC Codes

We further compare the performance of BA-SC-CCs with that of spatially coupled repeater-convolutional-combiner (SC-RCC) codes and BMST codes. For this comparison, BA-SC-CCs based on C_1 from Sec. IV-C are employed. The SC-RCC code C_{SC-RCC} from [16], with a decoding latency of 15,628 bits, is selected as a benchmark. Additionally, doubly recursive BMST (DrBMST) codes from [65] are considered, as they have been shown to outperform the original BMST codes [13]. The evaluated DrBMST codes have decoding latencies of 6,000, 15,000, 20,000, and 30,000 bits, with corresponding underlying uncoupled repetition codes of block lengths 750, 1,874, 2,500, and 3,750, respectively. The decoding delay for all DrBMST codes is set to 7.

The simulation results are shown in Fig. 10, from which we have the following observations.

- At the considered decoding latencies, the BA-SC-CCs perform better than the DrBMST codes. For example, at the decoding latency of 6000 bits, the considered BA-SC-CC outperforms the comparable DrBMST code by about 0.24 dB at $BER = 10^{-5}$.
- At the decoding latency of about 15,000 bits, the considered BA-SC-CC outperforms the comparable SC-RCC by about 0.2 dB at $BER = 10^{-6}$.

E. Comparisons With Braided Convolutional Codes

In this subsection, we compare the performance and the complexity of BA-SC-CC with those of the sparsely braided convolutional code. The considered decoding latency is 48,000 bits. We simulate a BA-SC-CC based on C_1 [6860, 3428, 2]. The decoding delay is $d = 6$. The SBC code for comparison is obtained by puncturing the SBC mother code [11] defined by two identical 4-state component encoders whose generator matrix is given by

$$G_4(D) = \begin{bmatrix} 1 & 0 & \frac{1}{1+D+D^2} \\ 0 & 1 & \frac{1+D+D^2}{1+D+D^2} \end{bmatrix}. \quad (17)$$

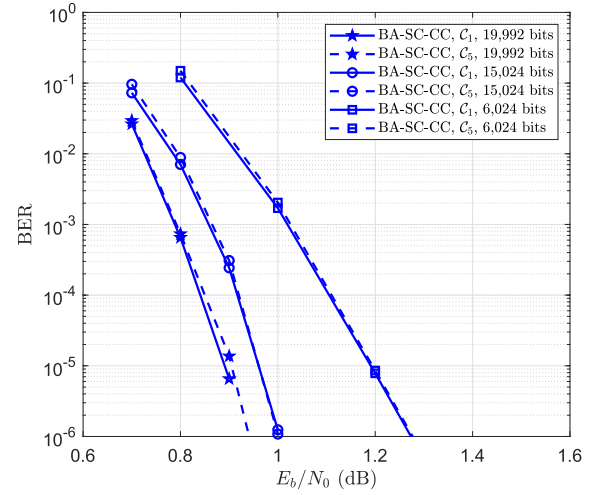


Fig. 11. Performance comparisons of BA-SC-CCs based on feedforward and recursive convolutional codes under the constraint of equal decoding latencies. These codes have comparable coding rates.

For this SBC code, the block permutor size corresponding to a decoding latency of 48,000 bits is 8,000. Decoding is performed using the sliding window decoder with uniform schedule [11], where the vertical and horizontal iteration numbers are set to 1 and 20, respectively. The decoding window size is set to $W = 3$, as shown in [11] to be optimal for finite-length SBCs. The simulation results are shown in Fig. 10, from which we observe that the considered BA-SC-CC performs 0.55 dB away from the channel capacity at bit error rate (BER) 2.0×10^{-5} and performs as well as the comparable SBC.

For each vertical iteration of the SBC code, the BCJR algorithm is executed twice. With the vertical iteration number set to 1, the SBC code based on $G_4(D)$ has twice the number of edges per decoding window compared to the BA-SC-CC based on C_1 . Simulation results show that, at a BER of 2×10^{-5} , the SBC code with an entropy-based stopping rule requires an average of 4.50 iterations. In contrast, the BA-SC-CC requires an average of only 2.01 iterations. Thus, the BA-SC-CC exhibits lower decoding complexity than the SBC code.

Remark 5: Due to the use of the BCJR algorithm in the sliding window decoder, the decoding complexity of BA-SC-CCs is typically higher than those of SC-LDPC codes and conventional BMST codes based on simple component codes.

Remark 6: Feedforward convolutional codes can also be used as the underlying uncoupled codes. For example, we have simulated a BA-SC-CC based on the feedforward convolutional code C_5 , defined by

$$G_5(D) = [1 + D^2 \quad 1 + D + D^2].$$

Both C_5 and C_1 have a free distance of 5. The considered decoding latencies include 6,000, 15,000, and 20,000 bits. Simulation results, shown in Fig. 11, indicate that these two BA-SC-CCs achieve comparable performance.

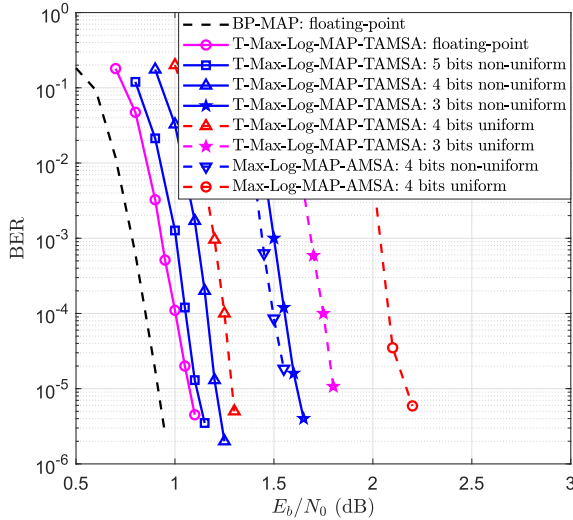


Fig. 12. Performances of a BA-SC-CC when decoded with different decoders.

V. A QUANTIZED THRESHOLD-BASED DECODER FOR BA-SC-CCs

We have shown that, for the chosen parameters, the BA-SC-CCs perform better than the regular SC-LDPC codes and the SC-RCCs. In the simulations in Sec. IV, the convolutional codes are decoded with the trellis-based maximum *a posteriori* (MAP) decoder, which however is not suitable for hardware implementation. From the implementation point of view, the low-complexity Max-Log-MAP algorithm is preferred to decode the CCs. In this section, we present a low-complexity quantized decoder for CCs and BA-SC-CCs.

An attractive feature of the BA-SC-CCs is that the degrees of the nodes in their normal graphs are at most three. The low-complexity quantized decoder operates iteratively on a subgraph of W layers of the normal graph of the BA-SC-CCs. To design such a decoder, we need to specify the decoding operations of the constraint nodes CC, $=$, Π , and $+$.

A. A Threshold-Based Decoder for BA-SC-CCs

In this subsection, we present the threshold-based decoder for the BA-SC-CCs with floating-point operations. Particularly, we present a threshold-based Max-Log-MAP decoder for CCs. The quantized counterpart of the threshold-based decoder will be presented in the next subsection.

1) *The Threshold-Based Max-Log-MAP Algorithm:* The nodes CC can be decoded with the Max-Log-MAP decoder. The operations of the Max-Log-MAP decoder include addition, comparison, and subtraction. In the following, we present a variant of the Max-Log-MAP decoder, the threshold-based Max-Log-MAP (T-Max-Log-MAP) decoder. For simplicity, we take the 1-input and n -output CC with constraint length v for illustration.

Let $L_{\text{in}}(i)$ denote the input LLR corresponding to the i -th coded bit and $L_{\text{app}}(i)$ denote the ratio of the *a posteriori* probability (APP) of the i -th coded bit in log domain when decoded with the Max-Log-MAP decoder. In the traditional

scaled Max-Log-MAP decoder, the extrinsic output $L_{\text{ex}}(i)$ for the i -th coded bit is given as

$$L_{\text{ex}}(i) = \lambda(L_{\text{app}}(i) - L_{\text{in}}(i)),$$

where $\lambda \in \mathbb{R}$ is the scaling factor. In the proposed threshold-based Max-Log-MAP (T-Max-Log-MAP) decoder, the extrinsic output for the i -th coded bit is computed as

$$L_{\text{ex}}(i) = \begin{cases} \lambda(L_{\text{app}}(i) - L_{\text{in}}(i)), & \text{if } \tau_0 < |L_{\text{app}}(i) - L_{\text{in}}(i)| < \tau_1; \\ L_{\text{app}}(i) - L_{\text{in}}(i), & \text{otherwise.} \end{cases} \quad (18)$$

In (18), τ_0 and τ_1 are two real numbers with $0 \leq \tau_0 < \tau_1$. It can be seen that, different from the traditional Max-Log-MAP decoder, the scaling operation in the T-Max-Log-MAP decoder is effective only when the magnitude of the extrinsic output is within the interval $[\tau_0, \tau_1]$. It is worth noting that the T-Max-Log-MAP decoder is not limited to BA-SC-CCs and can also be applied to the iterative decoding of other convolutional code-based schemes.

2) *The Threshold-Based Decoding at the Check Nodes:* Similar to LDPC codes, the min operation can be used for the nodes $+$. The threshold-based attenuated min-sum algorithm (TAMSA) in [43] and [66] can be employed in the proposed decoder. For completeness, we present the details of the threshold-based min operation in [66] in the following.

In BA-SC-CCs, the degrees of the nodes $+$ are at most three. Let L_1 and L_2 denote the two input LLRs of a given check node. In the scaled min-sum decoder of the LDPC codes, the extrinsic output L_{ex} of the node $+$ is computed as

$$L_{\text{ex}} = \lambda \cdot \text{sign}(L_1)\text{sign}(L_2) \cdot \min\{|L_1|, |L_2|\}, \quad (19)$$

where $\lambda \in \mathbb{R}$ is the scaling factor.

Similar to the T-Max-Log-MAP decoder, thresholds may be introduced to modify (19), resulting in the threshold-based min operation whose extrinsic output is given as

$$L_{\text{ex}} = \begin{cases} \lambda \cdot \text{sign}(L_1)\text{sign}(L_2) \cdot \min\{|L_1|, |L_2|\}, & \text{if } \tau_0 < \min\{|L_1|, |L_2|\} < \tau_1, \\ \text{sign}(L_1)\text{sign}(L_2) \cdot \min\{|L_1|, |L_2|\}, & \text{otherwise,} \end{cases} \quad (20)$$

where τ_0 and τ_1 are two real numbers with $0 < \tau_0 < \tau_1$.

B. Quantized Decoder for BA-SC-CCs

The decoder based on the T-Max-Log-MAP decoder and TAMSA are referred to as T-Max-Log-MAP-TAMSA in this paper. To reduce the implementation cost, the messages exchanged in T-Max-Log-MAP-TAMSA are typically quantized for a given bit width. The quantized decoder admits lower storage complexity and computation complexity than those of the floating-point decoders, but performs slightly worse. We propose to use non-uniform quantization (NUQ) for the decoding of BA-SC-CCs in this paper.

TABLE IV

THE QUANTIZER AND THE LOOK-UP TABLE FOR $\mathcal{F}$ WITH 3 BITS
QUANTIZATION: $\tau_0 = 1.91$ AND $\tau_1 = 3.27$

R_i	s_i	Q	$\mathcal{F}$
$(-\infty, -3.27]$	-4.75	-4	-4
$(-3.27, -1.91]$	-2.25	-3	-2
$(-1.91, -0.93]$	-1.37	-2	-2
$(-0.93, 0]$	-0.45	-1	-1
$(0, 0.93]$	0.45	0	0
$(0.93, 1.91]$	1.37	1	1
$(1.91, 3.27]$	2.25	2	1
$(3.27, \infty)$	4.75	3	3

1) *Non-Uniform Quantization:* Let $\mathbf{y} = (y_0, y_1, \dots, y_{(L+T)N-1})$ denote the received sequence from the channel. Then $\mathbf{y}$ is quantized with the scalar quantizer defined by $\mathbf{s} = (s_0, s_1, \dots, s_{J-1})$, where $s_0 < s_1 < \dots < s_{J-1}$. The interval quantized to the point s_i is $R_i = (r_i, r_{i+1}]$, where $r_0 = -\infty$ and $r_J = +\infty$. The mutual information between the signal message $\mathbf{X}$ and the quantized output is computed as,

$$I(X; S) = \frac{1}{2} \sum_{x=\pm 1} \sum_{s=s_0}^{s_{J-1}} P(s | X = x) \times \log_2 \frac{2P(s | X = x)}{P(s | X = -1) + P(s | X = +1)} \quad (21)$$

where

$$P(S = s_i | X = x) = \int_{y \in R_i} P(y | X = x) dy \quad (22)$$

In this paper, the method in [67] is used to construct the non-uniform quantizer. For completeness, we briefly introduce this method in the following. First, we partition the real line into a large number of basic intervals of equal length, except the leftmost one and the rightmost one. These basic intervals form the quantization interval of the 0-bit quantizer. Second, to construct the ℓ -bits quantizer with the $(\ell-1)$ -bits quantizer, we partition each quantization interval of the $(\ell-1)$ -bits quantizer into two quantization intervals. For each quantization interval, the partition resulting in the largest mutual information is selected. By repeating the above procedure, we can obtain a set of hierarchical quantizer.

2) *The Threshold-Based Decoder With Non-Uniform Quantization:* In the proposed decoder, the messages exchanged are quantized with a fixed non-uniform quantizer. We point out that when incorporating non-uniform quantization into T-Max-Log-MAP-TAMSA, the scaling factor λ may not work when the number of bits for quantization is small. In the following, we introduce an operator $\mathcal{F}$ to mimic the functions of the scaling and quantization.

Specifically, the operator $\mathcal{F}(x)$ first quantizes its argument x with the non-uniform quantizer and then takes the neighbouring quantization point of the quantization output with smaller LLR magnitude as the output. Hence, in the quantized

TABLE V

THE QUANTIZER AND THE LOOK-UP TABLE FOR $\mathcal{F}$ WITH 4 BITS QUANTIZATION: $\tau_0 = 0.90$ AND $\tau_1 = 3.22$

R_i	s_i	Q	$\mathcal{F}$
$(-\infty, -4.26]$	-4.79	-8	-8
$(-4.26, -3.22]$	-3.58	-7	-7
$(-3.22, -2.48]$	-2.78	-6	-5
$(-2.48, -1.89]$	-2.14	-5	-4
$(-1.89, -1.45]$	-1.59	-4	-3
$(-1.45, -0.90]$	-1.10	-3	-2
$(-0.90, -0.47]$	-0.65	-2	-2
$(-0.47, 0]$	-0.21	-1	-1
$(0, 0.47]$	0.21	0	0
$(0.47, 0.90]$	0.65	1	1
$(0.90, 1.45]$	1.10	2	1
$(1.45, 1.89]$	1.59	3	2
$(1.89, 2.48]$	2.14	4	3
$(2.48, 3.22]$	2.78	5	4
$(3.22, 4.26]$	3.58	6	6
$(4.26, \infty)$	4.79	7	7

T-Max-Log-MAP-TAMSA, the extrinsic output L_{ex} of the node + is given as

$$L_{\text{ex}} = \begin{cases} \mathcal{F} \{ \text{sign}(L_1) \text{sign}(L_2) \cdot \min[|L_1|, |L_2|] \}, & \text{if } \tau_0 < \min\{|L_1|, |L_2|\} < \tau_1; \\ \text{sign}(L_1) \text{sign}(L_2) \cdot \min[|L_1|, |L_2|], & \text{otherwise.} \end{cases} \quad (23)$$

and the extrinsic output of the i -th coded bit of the CC is given as

$$L_{\text{ex}}(i) = \begin{cases} \mathcal{F}(L_{\text{app}}(i) - L_{\text{in}}(i)), & \text{if } \tau_0 < |L_{\text{app}}(i) - L_{\text{in}}(i)| < \tau_1; \\ L_{\text{app}}(i) - L_{\text{in}}(i), & \text{otherwise.} \end{cases} \quad (24)$$

LUTs are employed to implement the proposed non-uniform quantization and enforce the condition $\tau_0 < |L_{\text{app}}(i) - L_{\text{in}}(i)| < \tau_1$. We present in Table IV a 3-bits quantizer Q and the look-up table (LUT) for the operator $\mathcal{F}$ [68], where $\tau_0 = 1.91$ and $\tau_1 = 3.27$. It can be seen that only the quantization outputs “-3” and “2” are scaled. The LUTs and the corresponding thresholds for 4 and 5 bits non-uniform quantizers are given in Table V and Table VI, respectively. Note that the numbers in boldface denote the quantization outputs which are scaled.

We use an example to illustrate the enforcement of the condition $\tau_0 < |L_{\text{app}}(i) - L_{\text{in}}(i)| < \tau_1$. In the case of 4-bits quantization shown in Table V, messages with magnitudes in the interval $(0.9, 3.22]$ are scaled. That is, $\tau_0 = 0.9$ and $\tau_1 = 3.22$. As an illustration, an LLR value of 1.6 is first quantized to 3. Since it lies within the scaling region, the final output after scaling is $\mathcal{F}(1.6) = 2$, which corresponds to the quantization interval $(0.9, 1.45]$.

The proposed T-Max-Log-MAP decoder operates on the trellis graph of the terminated convolutional code, i.e., the

TABLE VI
THE QUANTIZER AND THE LOOK-UP TABLE FOR $\mathcal{F}$ WITH 5 BITS
QUANTIZATION: $\tau_0 = 0.63$ AND $\tau_1 = 3.48$

R_i	s_i	Q	$\mathcal{F}$
$(-\infty, -4.67]$	-5.08	-16	-16
$(-4.67, -3.99]$	-4.30	-15	-15
$(-3.99, -3.48]$	-3.72	-14	-14
$(-3.48, -3.05]$	-3.25	-13	-12
$(-3.05, -2.68]$	-2.86	-12	-11
$(-2.68, -2.36]$	-2.52	-11	-10
$(-2.36, -2.06]$	-2.21	-10	-9
$(-2.06, -1.79]$	-1.92	-9	-8
$(-1.79, -1.53]$	-1.66	-8	-7
$(-1.53, -1.29]$	-1.41	-7	-6
$(-1.29, -1.06]$	-1.17	-6	-5
$(-1.06, -0.84]$	-0.95	-5	-4
$(-0.84, -0.63]$	-0.73	-4	-4
$(-0.63, -0.42]$	-0.52	-3	-3
$(-0.42, -0.21]$	-0.31	-2	-2
$(-0.21, 0]$	-0.10	-1	-1
$(0, 0.21]$	0.10	0	0
$(0.21, 0.42]$	0.31	1	1
$(0.42, 0.63]$	0.52	2	2
$(0.63, 0.84]$	0.73	3	3
$(0.84, 1.06]$	0.95	4	3
$(1.06, 1.29]$	1.17	5	4
$(1.29, 1.53]$	1.41	6	5
$(1.53, 1.79]$	1.66	7	6
$(1.79, 2.06]$	1.92	8	7
$(2.06, 2.36]$	2.21	9	8
$(2.36, 2.68]$	2.52	10	9
$(2.68, 3.05]$	2.86	11	10
$(3.05, 3.48]$	3.25	12	11
$(3.48, 3.99]$	3.72	13	13
$(3.99, 4.67]$	4.30	14	14
$(4.67, \infty)$	5.08	15	15

same state-transition structure used in BCJR decoder. However, unlike the conventional BCJR decoder, which performs exact forward-backward recursions, the T-Max-Log-MAP decoder adopts a threshold-based message updating rule within the Max-Log-MAP framework. This modification regulates message magnitudes during the recursion while preserving the standard trellis processing structure. Threshold-based message scaling and non-uniform quantization have been considered previously in TAMSAs. This work develops a threshold-based Max-Log-MAP decoder specifically for convolutional codes, where the threshold-controlled message update is embedded directly into the trellis-based recursion of Max-Log-MAP decoder [52].

For fixed-point implementations, the proposed quantized T-Max-Log-MAP decoder has the same implementation cost as the conventional quantized Max-Log-MAP decoder when LUTs are employed. Specifically, the thresholding and scaling operations can be fully incorporated into the mapping of

LUT, and therefore do not introduce additional memory and computational requirements beyond those already required by the quantized Max-Log-MAP decoder.

C. Simulation Results and Complexity Analysis

We present in Fig. 12 the BERs of a BA-SC-CC when decoded with the quantized T-Max-Log-MAP-TAMSA under different quantization schemes. Both uniform quantization and non-uniform quantization are considered. The terminated convolutional code $\mathcal{C}_1[2604, 1300, 2]$ is taken as the basic code. For comparison, we also present in Fig. 12 the error performances of this BA-SC-CC when decoded with different decoders. For the non-quantized T-Max-Log-MAP-TAMSA, the parameters are set to $\tau_0 = 0.75$, $\tau_1 = 4$, and $\lambda = 0.75$. For ℓ -bit uniform quantization, the interval $[-5, 5]$ is uniformly divided into 2^ℓ levels, resulting in a quantization step size of $\Delta = 5/2^{\ell-1}$. The corresponding thresholding parameters are selected as follows. For $\ell = 3$, we set $\tau_0 = 3.125$ and $\tau_1 = 4.375$. For $\ell = 4$, we set $\tau_0 = 1.563$ and $\tau_1 = 4.063$. For $\ell = 5$, we set $\tau_0 = 1.094$ and $\tau_1 = 4.219$. From Fig. 12, we have the following observations.

- 1) The T-Max-Log-MAP-TAMSA with 5 bits non-uniform quantization is sufficient to approach the performance of threshold-based decoder with floating-point operations.
- 2) For the proposed decoder, 4 bits non-uniform quantization performs 0.15 dB better than the 4 bits uniform quantization and the 3 bits non-uniform quantization performs 0.2 dB better than the 3 bits uniform quantization.
- 3) The T-Max-Log-MAP-TAMSA with 4 bits non-uniform quantization performs 0.3 dB better than the Max-Log-MAP-TAMSA with 4 bits non-uniform quantization and 0.5 dB better than the Max-Log-MAP-TAMSA with 4 bits uniform quantization. These results confirm the effectiveness of the proposed threshold-based decoder. Note that, in Max-Log-MAP-TAMSA, the CCs are decoded with the scaled Max-Log-MAP decoder and the check nodes are decoded with the AMSA.

For iterative decoders, decoding complexity is determined by both the computational cost per iteration and the average number of iterations required. In general, non-quantized decoders incur higher complexity than quantized decoders. For quantized decoders, the per-iteration complexity increases with the number of quantization levels, but remains comparable between threshold-based and conventional decoders when the same quantization resolution is used. This suggests that the overall decoding complexity is primarily governed by the average number of iterations. We consider a concrete example with 4 bits quantization for illustration. To achieve a BER of 10^{-5} , the average iteration numbers of the T-Max-Log-MAP-TAMSA with non-uniform and uniform quantization are both approximately 2.7, demonstrating that these two decoding schemes yield nearly identical decoding complexity.

VI. CONCLUSION

In this paper, we introduced the BA-SC-CCs and analyzed their asymptotic thresholds and error propagation

phenomenon. The analytical results showed that, with appropriate component convolutional codes, the decoding thresholds of BA-SC-CCs with a decoding delay of six were only 0.1 dB away from the channel capacities. We then provided a lower bound on the free distance of the BA-SC-CC. To reduce the implementation cost, we introduced a threshold-based decoder for BA-SC-CCs based on non-uniform quantization and the threshold-based Max-Log-MAP decoder. Comprehensive comparisons showed that, under equal decoding latency, BA-SC-CCs perform as well as the sparsely braided convolutional codes and outperform spatially coupled repeater-combiner-convolutional codes and regular SC-LDPC codes. The comparisons also showed that, for low-resolution quantization, performance gains as large as 0.2 dB can be obtained by the threshold-based decoder. We point out that analyzing the ensemble behavior of the free distance of BA-SC-CCs is an important and promising direction.

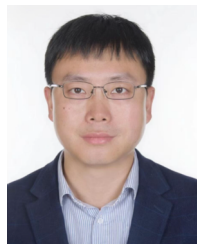
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